Weather Data Analysis Report

1. Introduction

Weather analysis plays an important role in understanding climate patterns and environmental conditions. By analyzing historical weather data, we can identify relationships between variables such as temperature, humidity, wind speed, and pressure.

This project focuses on analyzing weather data using **Python, Pandas, Matplotlib, and Seaborn** to extract meaningful insights and visualize patterns.

2. Objective

The objectives of this project are:

- To analyze weather parameters like temperature, humidity, wind speed, and pressure
- To study the distribution of humidity
- To identify correlations between weather variables
- To find extreme weather conditions (highest and lowest temperature days)

3. Tools and Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook

4. Dataset Description

The dataset contains historical weather data with features such as:

Column	Description
Date	Date and Time of observationm
Temperature	Temperature value
Humidity	Humidity percentage
wind_ speed	Wind speed
Pressure	Atmospheric pressure
_conds	Weather Condition
_dewptm	Dew Point
_fog, _rain, _snow	Weather Indicators

5. Data Loading

The dataset was successfully loaded into the environment using Pandas.

df = pd.read_csv("weather_data.csv")
print("Data loaded successfully")

6. DataAnalysis and Visualization

6.1 Humidity Distribution

sns.histplot(df['humidity'], bins=30)
plt.title("Humidity Distribution")
plt.show()

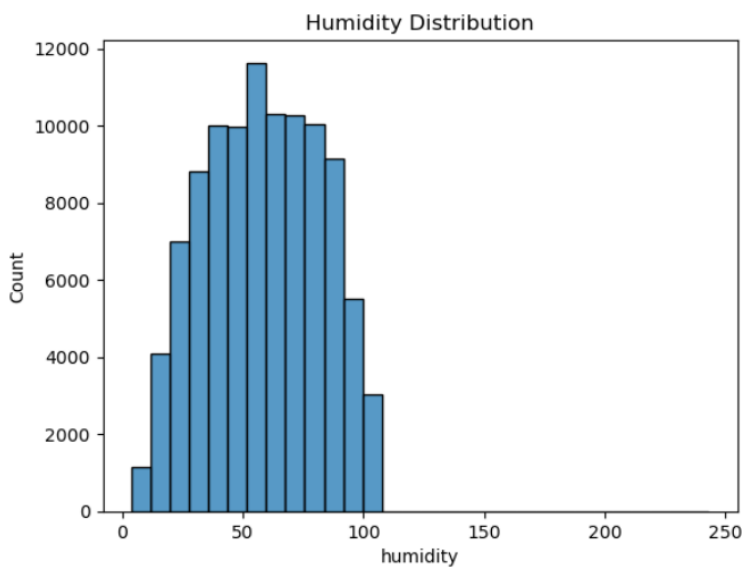

Figure 1: Humidity Distribution

Interpretation:

The histogram shows how humidity values are distributed. It helps understand whether humidity levels are mostly high, low, or evenly distributed.

6.2 Correlation Analysis

correlation = df[['temperature','humidity','wind_speed','pressure']].corr()
print(correlation)

Correlation Results

Variable	Relationship
Temperature vs Humidity	-0.589 (Strong Negative)
Temperature vs Wind Speed	0.214(Weak Positive)
Temperature vs Pressure	0.004(No Relation)

6.3 Heatmap Visualization

```
corr = df[['temperature','humidity','pressure','wind_speed']].corr()
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title("Weather Correlation")
plt.show()
```

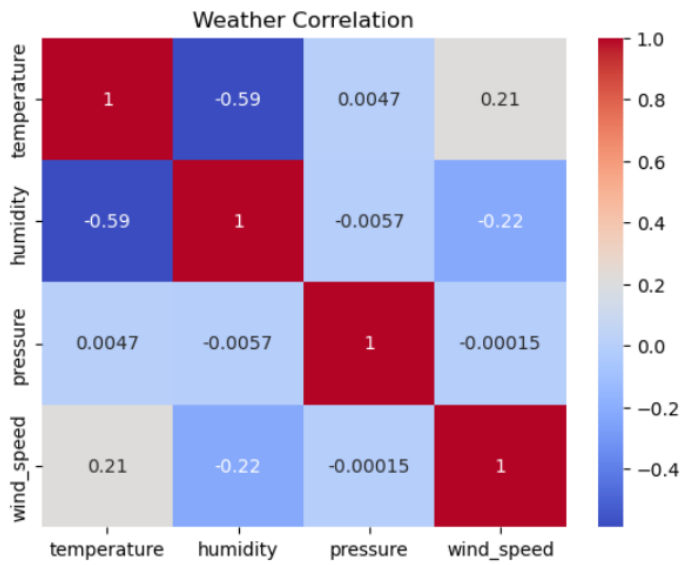

Figure 2: Correlation Heatmap

Interpretation:

- Temperature and humidity are negatively correlated.
- Wind speed has a slight positive relation with temperature.
- Pressure shows almost no relationship with other variables.

6.4 Highest Temperature Day

```
print(df.loc[df["temperature"].idxmax()])
```

Result:

- Date: 2015-06-24
- Temperature: 90.0
- Condition: Patches of Fog
- Humidity: 4%

Interpretation:
This was the hottest recorded day in the dataset with very low humidity, indicating dry and hot weather conditions.

6.5 Lowest Temperature Day

print(df.loc[df["temperature"].idxmin()])

Result:

- **Date: 2006-01-08**
- **Temperature: 1.0**
- **Condition: Clear**
- **Humidity: 75%**

Interpretation:
This was the coldest day, with high humidity and clear weather conditions.

7. Key Insights

- Temperature and humidity have a **strong negative relationship**
- Higher temperature usually corresponds to lower humidity
- Wind speed has a weak effect on temperature
- Pressure has almost no correlation with other variables
- Extreme weather days provide insights into climate variability

8. Conclusion

This project successfully analyzed weather data and identified relationships between key weather parameters. The findings show that temperature and humidity are inversely related, while other variables like pressure have minimal impact.
Such analysis can be useful for:

- Climate research
- Weather forecasting
- Environmental studies

9. Future Improvements

- Add time-series analysis for trends over years
- Use machine learning for weather prediction
- Build an interactive dashboard using Power BI or Tableau